

Madisyn Messmore

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Ph.D. Candidate in Biomedical engineering, with a Master of Science in Biomedical Engineering under the Biofluids track and an Honors Bachelor of Science degree in Mechanical Engineering from UCF. My research interests include biofluid mechanics of the cardiovascular system, computational fluid dynamics, medical image processing, and machine learning for clinical applications.

EDUCATION

University of Central Florida (2022 - 2026)

- Ph.D. in Biomedical Engineering
- 3.9 Cumulative GPA

University of Central Florida (2022 - 2024)

- Master of Science in Biomedical Engineering - Biofluid Mechanics Focus
- 3.938 Cumulative GPA

University of Central Florida (2018 - 2022)

- Honors Bachelor of Science in Mechanical Engineering (BSME)
- Member of the Burnett Honors College at UCF
- 3.38 Cumulative GPA

Niceville Senior High School (2014-2018)

- 4.625 Weighted GPA; 3.867 Unweighted GPA, 33 ACT
- National Merit Commended Scholar and National AP Scholar

RESEARCH AND WORK EXPERIENCE

- **Graduate Research Assistant at the Computational Mechanics Lab, UCF** (May 2022 - Present)
 - Conducting 3D Star-CCM+ unsteady, internal flow simulations to model hemodynamics of the heart.
 - Completed a coupled unsteady multiphase 2D CFD model of the heart with moving mesh to replicate real patient heart motion.
 - Wrote MATLAB code to solve a lumped parameter flow model using linear electrical circuit principles.
 - Derived oxygen transfer equations from mass conservation principle and applied formal optimization methods for system optimization.
 - Coupled CFD simulations, lumped parameter flow model, and oxygen transfer model in an iterative process by using high-powered computing.
 - Created unsteady 3D CFD model of aortopulmonary collateral vessels with multiphase model to simulate occlusion via PVA beads.
 - Created 3D model of aortopulmonary collateral vessels from angiographic image sets for CFD.
 - Collaborator on in-vitro benchtop model for the optimization of cannula angle in LVAD patients to mitigate stroke incidence. Project lead for a related investigation into hemolysis in LVADs.
 - Hold a mentorship and managerial role in the lab for maintaining lab standards and assisting in the growth of undergraduate researcher careers.
 - **IRB Approved Study with Orlando Health & UCF** for "Development and Validation of a Computational Fluid Dynamic Model of Dextro-transposition of the Great Arteries"
 - **IRB Approved Study with Orlando Health & UCF** for "Pre-Operative Clinical Course of Newborns With Transposition of The Great Arteries"
- **Graduate Teaching Assistant (2024 – Present)**
 - Assist with all technical needs of the graduate anatomy courses, Biostructures I & II.
 - Created a class project using CT scans of cadavers and 3Dslicer to better students understanding of cardiovascular anatomy in 3D space.
 - Interface with IT, the cadaver lab, and course professor to ensure smooth transition of cadaver medical scans to students in a usable format.
- **Arnold Palmer Children's Hospital Observership** (2022 - Present)
 - Pediatric Cardiothoracic Surgery Shadow under Dr. William Decampoli at Arnold Palmer Children's Hospital to increase understanding of ongoing research projects.

- Shadowed echocardiography technicians during the capture of imaging related to TGA study.
- **Undergraduate Research at the Biomaterials & Nanomedicine Lab, UCF** (Jan 2020 - May 2022)
 - Designed biodegradable bone implants using Solidworks (e.g., scaffolds, screws, rods).
 - Performed Ansys Mechanical FEA to demonstrate sufficient design strength of implants.
 - Implemented Ansys Fluent CFD to calculate internal pressures of a novel biofiltration system.
- **UCF Honors Undergraduate Thesis (May 2021 – May 2022)**
 - Wrote and defended: “[Design and Development of Medical Devices for Multifaceted Applications](#)”.
- **Undergraduate Teaching Assistant** (May 2021-May 2022)
 - Teaching assistant for upper-level engineering course, Modeling Methods (EML 3034C). Tasked with teaching lab sections on application of mathematical theory with programming in MATLAB/Octave GNU, as well as serving as a mentor for peers.
- **Howe Industries Independent Study Remote Student** (Feb - May 2021, Scottsdale, AZ)
 - Analysis of orbital mechanics, basic principles of solar energy, and propulsion techniques to develop novel efficient technologies using MATLAB, Python, and STK
- **Limbitless Solutions Engineering Scholar** (Fall 2019), **Administration Scholar** (Summer 2019)
 - Experience with CAD in Fusion360, Arduino programming, additive manufacturing techniques, and STEM education initiatives

GRANTS, AWARDS, & CERTIFICATIONS

- UCF Graduate Presentation Fellowship (2025)
- Co-authored Saving Tiny Hearts Society Research Grant application (2025)
- Applied to NIH Predoctoral Fellowship 2024 - received impact score of 36, not awarded
- Applied to AHA Predoctoral Fellowship 2023 - not awarded
- University of Central Florida's ORCGS Doctoral Fellowship Recipient (2022-2023)
- Walt Disney World Co. Design and Engineering Scholarship (2022-23)
- Allyn M. Stearman Research Scholarship (2022)
- Provost Scholar at the University of Central Florida (2018-2022)
- President's Honor Roll (2020)
- Dean's List in the College of Biomedical Sciences (2019), College of Engineering (2020-21)
- Several certifications in Web Development, Microsoft Word, Excel, and PowerPoint
- Certified SOLIDWORKS Associate in Mechanical Design (CSWA)

PUBLICATIONS

- Messmore, M., DeCampi, W. & Kassab, A. *Computational Model for Predicting Optimal Clinical Intervention in Pre-Operative Neonates with Transposition of the Great Arteries*. Cardiovasc Eng Tech (2026).
<https://doi.org/10.1007/s13239-026-00839-9>
- Messmore, M., DeCampi, W., Kassab, A. J., Prather, R., & Castillo Arceo, D. A. (2024). *Cilia and nodal flow in asymmetry: An engineering perspective*. Critical Reviews in Biomedical Engineering.
<https://doi.org/10.1615/critrevbiomedeng.2024051678>
- Razavi, M., Aboutalebianarki N., Messmore, M., Shar, A., & Misiti, K. (2022). “Musculoskeletal Tissue-On-a-Chip” in Principles of Human Organs-on-Chips, S.I.: WOODHEAD PUBLISHING UK.

In Preparation:

- Messmore, M., DeCampi, W. & Kassab, A. *Two-Dimensional Analysis of the Heart Ventricles for Dextro-Transposition of the Great Arteries*
- Messmore, M., Yishay, T., Santos, C., Mbizo, A., Ellerbe, A., Fellows, K., Castillo Arceo, D. A. *Hemolysis in Left Ventricular Assist Devices*

CONFERENCES

- UCF Biionix Cluster 4th Annual Research Symposium, Presenting Author, podium presentation titled “*Multiscale Modeling of Transposition of the Great Arteries*”
- Florida Fluids Symposium 2026, podium presentation titled “*Two-Dimensional Analysis of the Heart Ventricles for Dextro-Transposition of the Great Arteries*”

- ASEMFL Annual Meeting 2025, Presenting Author, poster titled “*Multiscale Modeling of Transposition of the Great Arteries*”
- ASME Summer Bioengineering 2025, Presenting Author, podium presentation titled “*Optimization of Pre-Operative Transposition of the Great Arteries Care Strategies Utilizing Mathematical Modeling*”
- Florida Fluids Symposium 2025, Presenting Author, podium presentation titled “*Computational Modeling of Artificial Flow Determining the Left-Right Axis in the Embryonic Node*”
- UCF Biionix Cluster 3rd Annual Research Symposium, podium presentation titled “*Updates on Hemodynamic Modeling of Transposition of the Great Arteries*”, May 2025
- CHOP Cardiology 2025, Presenting author, poster titled “*Mathematical Modelling Predicts the Relative Importance of Potentially Modifiable Parameters to Optimize Oxygenation in Transposition of the Great Arteries*”
- UCF Student Scholar Symposium 2024, Presenting author, poster titled “*An Investigation of Correlations Between Cardiovascular Pathology and Dementia Using Anatomical Donors*”
- UCF Student Scholar Symposium 2024, Assisting author, poster titled “*Left Ventricular Assist Devices: Thrombus Prevention and Controlled Migration*”
- UCF Biionix Cluster 3rd Annual Research Symposium, podium presentation titled “*Hemodynamic Modeling of Transposition of the Great Arteries*”, May 2024
- Florida Undergraduate Research Conference (FURC) 2022, Presenting author, poster titled “*Design and Development of Medical Devices for Multifaceted Applications*”
- UCF Student Scholar Symposium 2022, Presenting author, poster titled “*Design and Development of Medical Devices for Multifaceted Applications*”

LECTURES

UCF

- **May 2025:** Gauss Elimination & LU Decomposition; *EML 3034C - Modeling Methods in MAE*
- **May 2025:** Systems of Simultaneous Linear Equations; *EML 3034C*
- **July 2024:** Numerical Methods in Ordinary Differential Equations; *EML 3034C*
- **June 2024:** SOR, Matrix Norms, & Conditioning Numbers; *EML 3034C*
- **June 2024:** Non-linear Solvers, Newton-Raphson, & Steepest Descent; *EML 3034C*
- **June 2024:** Lagrange Interpolation and Cubic Splines; *EML 3034C*
- **June 2024:** Linear & Non-Linear Least-Squares; *EML 3034C*
- **June 2024:** Finite Differences & Gauss Quadratures; *EML 3034C*
- **June 2024:** Newton-Cotes, Adaptive Integration, and Romberg Integration; *EML 3034C*
- **June 2022:** Finite Differences and Gauss Quadratures; *EML 3034C*
- **July 2022:** Newton-Cotes, Adaptive Integration, & Romberg Integration; *EML 3034C*
- **October 2022:** Runge-Kutta Methods & Adaptive Time Steps; *EML 3034C*

ORGANIZATIONAL MEMBERSHIPS

- ASME
- BMES
- SWE
- IEEE

EXTRACURRICULARS AND PROJECTS

- **Continuous Improvement of Biomedical Engineering Education Project**
 - Member of a team of graduate Biomedical Engineering students at UCF working with faculty of the college of medicine and the college of engineering to improve the curriculum for BME students. This includes the creation of a 3D Slicer project, and surveys to optimize student experience.
- **Benchtop Verification of Optimal LVAD Cannula Angle to Prevent Stroke Risk**
 - Member of a long-standing group in collaboration with Embry-Riddle Aeronautical University to validate the CFD simulations of varying cannula angles to redirect thrombus migration.
- **Graduate Research Mentorship (FLEARN) (2024-Present)**

- Mentor to 2-3 undergraduate freshmen each academic year to promote women in STEM research. Developed mini-projects for them to explore the field as well as assist in authoring either a research poster for the UCF research symposium or a paper for publication.
- **UCF Honors Undergraduate Thesis** (May 2021-May 2022, UCF)
 - Honors Thesis Program in biomedical technology titled “[Design and Development of Medical Devices for Multifaceted Applications](#)”; involved individually writing and defending a thesis on biofiltration systems and the design of implantable biomedical devices.
- **ASME Human Powered Vehicle Competition: Design & Analysis Lead** (2021-2022)
 - Designed, built, and tested a recumbent trike with e-assist on a team for a final engineering design project as a part of the Senior Design coursework. Spearheaded frame design, ensuring it met manufacturing and design constraints, as well as performed all necessary FEA and CFD analyses. The team placed 6th in the [CDR competition](#) and 4th in the [Innovation Challenge](#)
- **Students for the Exploration and Development of Space (SEDS): Director of External Affairs** (2019-2020), **member** (2018-2022)
 - Advocated for diversity in space, maintained corporate relationships, organized events, and led branding efforts for one of the largest STEM student organizations at UCF
- **LimbitlessEDU Documentary, Engineering lead, Main Cast Member, and Voice Actor** (2019)
 - Starred in an educational documentary mini-series, specifically in an episode on 3D printing. Edited the engineering content, delivered performance, and assisted in production.
- **Limbitless Solutions Education Outreach Student Leader** (2019)
 - Volunteered for several educational outreach events, including an elementary school exhibit as well as working a booth at MegaCon in Orlando, demonstrating the arm’s technology
- **Camp Code Orlando, Camp Leader and Curriculum Designer** (Limbitless Solutions, 2019)
 - Designed and led a week-long summer camp to teach underprivileged high school students about programming, engineering, and manufacturing.
- **Varsity Golf Team** (2014-2018) and **Captain** (2016-2018)
 - State competitors three years in a row, State runner-up team in 2016-17 season and 2018-19 season. Placed 12th individually in the state in the 2018 FHSA Championship.

RELEVANT COURSEWORK

Graduate

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| ● Computational Fluid Dynamics (CFD) I | ● Continuum Biomechanics | ● Advanced Biomechanics |
| ● CFD II | ● Mechanics of Viscous Flow | ● Engineering Statistics |
| ● Biofluid Mechanics | ● Turbulent Flow | ● Mechanics of Biostructures I |
| ● Biomedical Modeling | ● Bioinstrumentation | ● Mechanics of Biostructures II |
| ● Applied & Computational Biofluid Mechanics | ● Topics in Biomedical Eng. | |
| | ● Nanoscale Bioengineering | |
| | ● Clinical Embryology | |

Undergraduate

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| ● Solid Mechanics | ● Principles of Electrical Engineering | ● Vibrations & Controls |
| ● Intro to Chemistry I & II | ● Honors Thesis | ● Microbiology |
| ● Intro to Programming with C | ● Modeling Methods | ● Tissue Engineering |
| ● Engineering Statics | ● Thermo of Mech. Systems | ● Physiological Psychology |
| ● Engineering Dynamics | ● Mech. Measurements | ● Honors Statistics |
| ● Honors Thermodynamics | ● Fluid Mechanics I | ● Statistics for Engineers |
| ● Structure & Prop of Materials | ● Heat Transfer I & II | ● Honors Diff. Equations |
| | | ● Calculus I, II, & III |

SKILLS, SOFTWARE, AND PROGRAMMING LANGUAGES

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|-------------------|--------------|----------------|
| ● Time Management | ● Leadership | ● Organization |
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- Scientific Writing in LaTeX
- Experimental Design
- 3D Printing
- Soldering
- STAR-CCM+
- SimVascular
- CRIMSON
- 3DSlicer
- Mimics
- Fusion 360
- Ansys Mechanical
- Ansys Fluent
- NI LabVIEW
- SolidWorks
- MATLAB
- Julia
- HTML
- CSS
- JavaScript
- Python
- C/C++
- Blender
- Formal Optimization
- Numerical Methods

TRAININGS

CITI Trainings

- Human Subjects Research- Group 1. Biomedical Research Investigators and Key Personnel
- Biomedical Responsible Conduct of Research
- Responsible Conduct of Research for Engineers
- Conflict of Interest
- Lab Animal Researcher/ Technicians/ Students
- GCP for Clinical Trials with Investigational Drugs and Biologics (ICH Focus)
- Communicating Research Findings
- Revised Common Rule
- Research and HIPAA Privacy Protections
- Researchers - Information Privacy & Security (IPS)
- Biomedical Research - Basic/Refresher
- CITI Health Information Privacy and Security (HIPS) for Clinical Investigators

Graduate Academic Integrity (RCR) Trainings

- Doing the Right Thing (2022)
- Ethical Decisions in Graduate School (2022)
- Intellectual Property Rights (2024)
- Ethics: Personal Integrity as a Graduate Student (2024)

COMMUNITY SERVICE

- Volunteer Judge for the UCF Senior Design Showcase (2024)
- Habitat for Humanity, participated in several build events and restore events (2021)
- Camp Connect volunteer through SEDS-UCF, teaching rocketry principles to students (2019-2020)
- Habitat for Humanity at UCF volunteer and Public Relations Director (2018-2019)